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Article

PFAS in Rural U.S. Well Water: Using Participatory Science to Identify and Communicate Results to Address Risks

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ABSTRACT: Drinking water contributes 1.2−61% of perfluoroalkyl and polyfluoroalkyl substances (PFAS) exposure, but little information is available on PFAS in private wells serving 17% of the U.S. population. Well users (n = 271) from four U.S. states collected tap water samples for analysis of 25 PFAS. Participants received results report comparing their water to PFAS health guidelines and recommending water filters when guidelines were exceeded. Follow-up surveys asked respondents what actions they took in response. To identify potential PFAS sources, we computed distances from households to PFAS sources recorded in EPA's PFAS Analytic Tools and other public databases and used spatial regression models to analyze relationships between these sources and water quality. PFAS were found in 15% of wells in the area with no known sources, some above a health guideline, and 53−88% of wells at sites with known sources. Total PFAS concentrations were significantly higher in wells closer to PFAS production facilities, Superfund sites, spill sites, and federal facilities. When recommended, 49% installed filters, while none did where water met the guidelines. Our study sheds insight into geographic variation in PFAS in private wells, the influence of potential PFAS sources on well water quality, and private well user decision-making in response to PFAS water quality information.



KEYWORDS: *data report-back, drinking water, environmental health, PFAS, private well, risk communication, spatial regression*

INTRODUCTION

Popularly known as "forever chemicals," perfluoroalkyl and polyfluoroalkyl substances (PFAS) are ubiquitous contaminants that persist in the environment, some of which also bioaccumulate. Because of these characteristics, households relying on private wells in areas where PFAS is manufactured, used, or discarded may be at risk for PFAS exposure through drinking water. PFAS are organic human-made compounds introduced in the 1940s and used in a wide variety of applications, including firefighting foam, nonstick cookware, water-repellent and stain-resistant fabrics, and many other consumer and industrial products.[1] Human health effects associated with PFAS exposure include immune system suppression, dyslipidemia, liver damage, thyroid disorders, and cancer.[1−4]

A recent study found that household tap water can contribute 1.2−61% of individual PFAS chemical concentrations observed in human plasma.[5] Recognizing the importance of drinking water as an exposure route, in April 2024, the U.S. EPA adopted National Primary Drinking Water Regulations and health-based Maximum Contaminant Level Goals (MCLGs) for six PFAS compounds: perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid (PFOS), perfluorononanoic acid (PFNA), hexafluoropropylene oxide dimer acid (HFPO−DA, commonly known as GenX), perfluorohexanesulfonic acid (PFHxS), and perfluorobutanesulfonic acid (PFBS).[6] The new regulations require all public water systems to test their water for these PFAS by 2027. If concentrations exceed the new standards, water utilities must take action to provide water that meets the standards (through treatment, source water protection, or a change in water source).

The new PFAS regulations do not apply to private water systems, defined as those serving fewer than 15 service connections or 25 people or those operating fewer than 60 days per year. An estimated 17% of the U.S. population relies on private systems, most of which draw their water from wells.[7] Recent investigations have detected PFAS in private wells in many locations. For example, two multistate U.S. Geological Survey studies (one including 269 and the other 49 private

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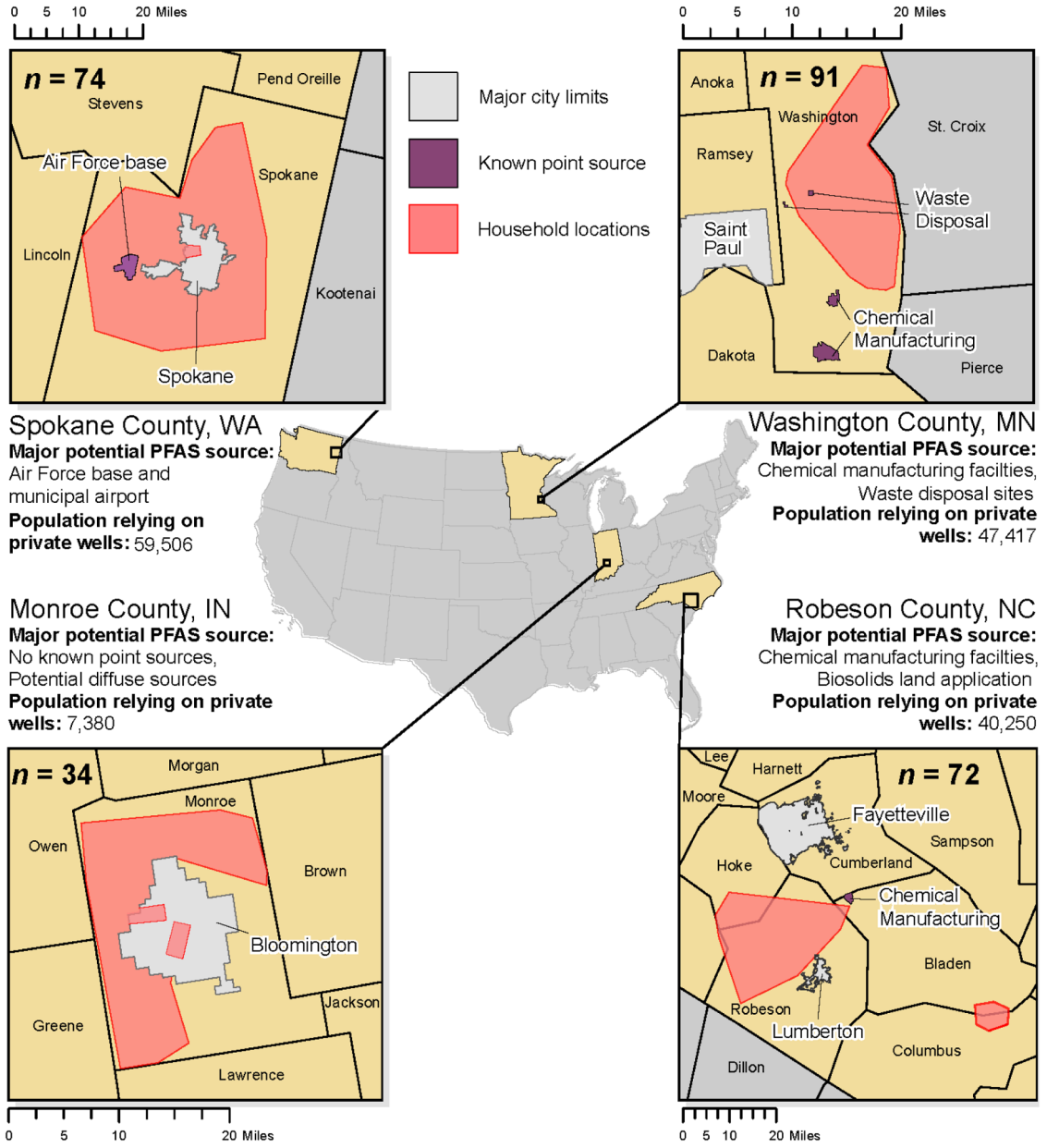


Figure 1. Four case study sites across the United States, potential PFAS sources at each site, and approximate locations of households relying on private wells recruited for participatory sampling.

wells) found at least one PFAS in 20% of wells.[8,9] A New Hampshire study found PFAS in 17−48% (depending on the PFAS type) of private wells,[10] while a Virginia study found at least one PFAS in 95% of private wells.[11]

With mounting evidence that PFAS can contaminate private wells, methods are needed to identify areas at risk. Due to general lack of awareness of the need for water quality monitoring and the high costs associated with PFAS testing, the vast majority of private well users lack information to guide decision-making about whether they should act to protect themselves and their families from PFAS in their tap water (for example, by ordering water tests to determine whether to purchase a filter capable of PFAS removal).[12,13] Public health departments in some states and counties have outreach programs to support private well owners and promote testing and can make use of recently developed resources like the PFAS in US Groundwater Interactive Dashboard.[14] However, these programs typically are understaffed and lack resources to support statewide testing.[15] The U.S. EPA launched its PFAS Analytic Tools in January of 2023, compiling data on potential PFAS sources across the nation,[16] but this database does not provide information about the likelihood of private well contamination arising from these sources or the distances over which they may influence contamination risk. In addition, due to the technical complexity of PFAS (including complicated-sounding names and many varieties), well owners and public health officials require guidance on interpreting analytical reports in cases where well water testing data are available.

Using a participatory science approach, this study addresses gaps in knowledge about risk factors for PFAS in private wells and methods for communicating information about PFAS to private well owners. The study engaged households relying on private wells in four geographic areas, three with different known PFAS sources and one with no known major sources.

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Using results from well water testing in these communities, publicly available data on known or potential PFAS sources, and participant surveys, we aimed to answer three questions:

1. How do levels and composition of PFAS in private wells vary across these four sites?
2. Which potential PFAS sources reported in EPA's PFAS Analytic Tools and other public databases are associated with PFAS occurrence in private wells, and over what distances do those sources influence PFAS concentrations?
3. How do well users respond when receiving results of PFAS water tests?

METHODS

Study Design. Four case study sites (Figure 1) were selected in distinct geographic regions to include several examples of known PFAS sources and one control site with no known PFAS releases. The Airway Heights community in Spokane County, WA, is near an air force base and municipal airport, where PFAS were found in the public water system municipal wells in 2017.[17] Washington County, MN, has areas of known groundwater and private well contamination from manufacturing and disposal of PFAS by a manufacturing facility.[18] Robeson County, NC, is adjacent to a chemical manufacturing facility that dispersed PFAS through stack emissions and effluent, affecting nearby private wells.[19] Lastly, Monroe County, IN, serves as a control site and is not home to any known PFAS point sources.

Eligible participants lived in one of the study counties and had a private well on their property. Participants were recruited via mail and phone using addresses obtained from local partners and a Community Advisory Board (Supporting Information (SI)). Participants completed an enrollment survey (SI Appendix 1) with questions about their well (year of construction, depth), use of water from the well (e.g., drinking, cooking, and watering garden), use of household water treatment, and household demographic information. The study was approved by the Institutional Review Board at North Carolina State University (IRB #25816).

Sample Collection and Laboratory Analysis. Enrolled participants received a sample collection kit in an insulated box with a free overnight return shipping label. The kits included two empty 250 mL polypropylene bottles containing ammonium acetate as a preservative and one prefilled bottle containing reagent-grade water as a field blank. Five percent of participants received additional empty bottles used as duplicates for quality assurance. All kits included cold packs, gloves, chain-of-custody documents, and written sample collection instructions (SI Appendix 2) with a phone number to call for help. Our team answered questions by phone, e-mail, and optional webinars to which all participants were invited. Participants flushed water for 4 min before collecting samples, poured the reagent-grade water in the prefilled bottle into the additional empty bottle, and returned the samples, cold packs, and chain-of-custody forms in the insulated box. Samples were collected between May 2021 and May 2022.

Samples were analyzed at RTI International's laboratory for 25 PFAS by liquid chromatography/tandem mass spectrometry (LC-MS/MS) using a SCIEX ExionLC AC, LC system, and SCIEX 7500 triple quadrupole MS/MS instrument according to EPA Method 533.19. A laboratory minimum reporting limit (MRL) of 0.888 to 2 ppt (Table S6.1) was determined for all analytes (see SI for Quality Assurance Methods). A field blank was collected alongside each field sample. Fifteen percent of field blanks contained one or more PFAS above one-third of the method reporting limit (MRL), including cases where participants self-reported mistakenly using tap water. About 2% of data points (142 of 6775) were affected by positive field blanks. Data quality flags were applied to 0.4% of data points (24 of 6775) where PFAS were detected in both the sample (above MRL) and its corresponding blank (above 1/3 MRL). All field sample results were retained for analysis.

Evaluation of Results Report-Back. A report-back template (SI Appendix 3) was developed in July 2022 in accordance with best practices[20−23] and the research team's previous experience.[24,25] These best practices included presenting results in both textual and graphical formats, clearly summarizing key takeaways in plain language (including specific ways to reduce exposures), and providing opportunities to talk directly with researchers about the results report (see SI for details).

A short postparticipation survey (SI Appendix 4) was distributed by e-mail or mail to 261 participants 3−12 months after receiving results reports. Results reports were undeliverable to 10 participants because they moved without leaving a forwarding address. The survey included questions about actions taken in response to the results; changes in perceptions of well water quality; perceived self-efficacy; the utility of the results reports, webinars, and meetings; overall satisfaction with the information provided.

Identification of Potential PFAS Sources and Neighborhood Demographic Information. Data on nine potential PFAS source categories were downloaded in January 2023 from EPA's PFAS Analytic Tools database and other public databases: PFAS spill sites, PFAS production or utilization facilities, Superfund sites, waste transfer sites handling PFAS, federal facilities with known or suspected PFAS, PFAS handling sites (including fire-training sites and airports), wastewater discharge locations, Toxics Release Inventory sites reporting PFAS use, and wastewater biosolids production sites (Table S2).[16] We removed duplicates within and across the data sets, as described in the Supporting Information.

Linear distances from each participant's address to the nearest source of each type were calculated. Median household income and population density of the block group in which each private well was located were obtained from the 2018 to 2022 American Community Survey.[26]

Statistical Analyses. Differences in the sum of 25 PFAS compounds (EPA 533; $\sum PFAS_{25}$) by site were assessed using nonparametric Kruskal−Wallis and Wilcoxon rank-sum tests. Kruskal−Wallis tests evaluated the overall significance of differences across the four study sites. Wilcoxon rank-sum tests evaluated pairwise differences in median concentrations between the sites.

Effects of participating in the study on decision-making (e.g., starting or stopping water use for drinking or cooking, purchasing bottled water, installing a filter) were analyzed via chi-square tests of proportions to assess differences among those receiving a recommendation to install a water filter compared to participants with lower PFAS concentrations and no recommendation to install a filter. In addition, chi-square tests of proportions or Fisher's exact tests for two proportions were performed to analyze differences in demographic

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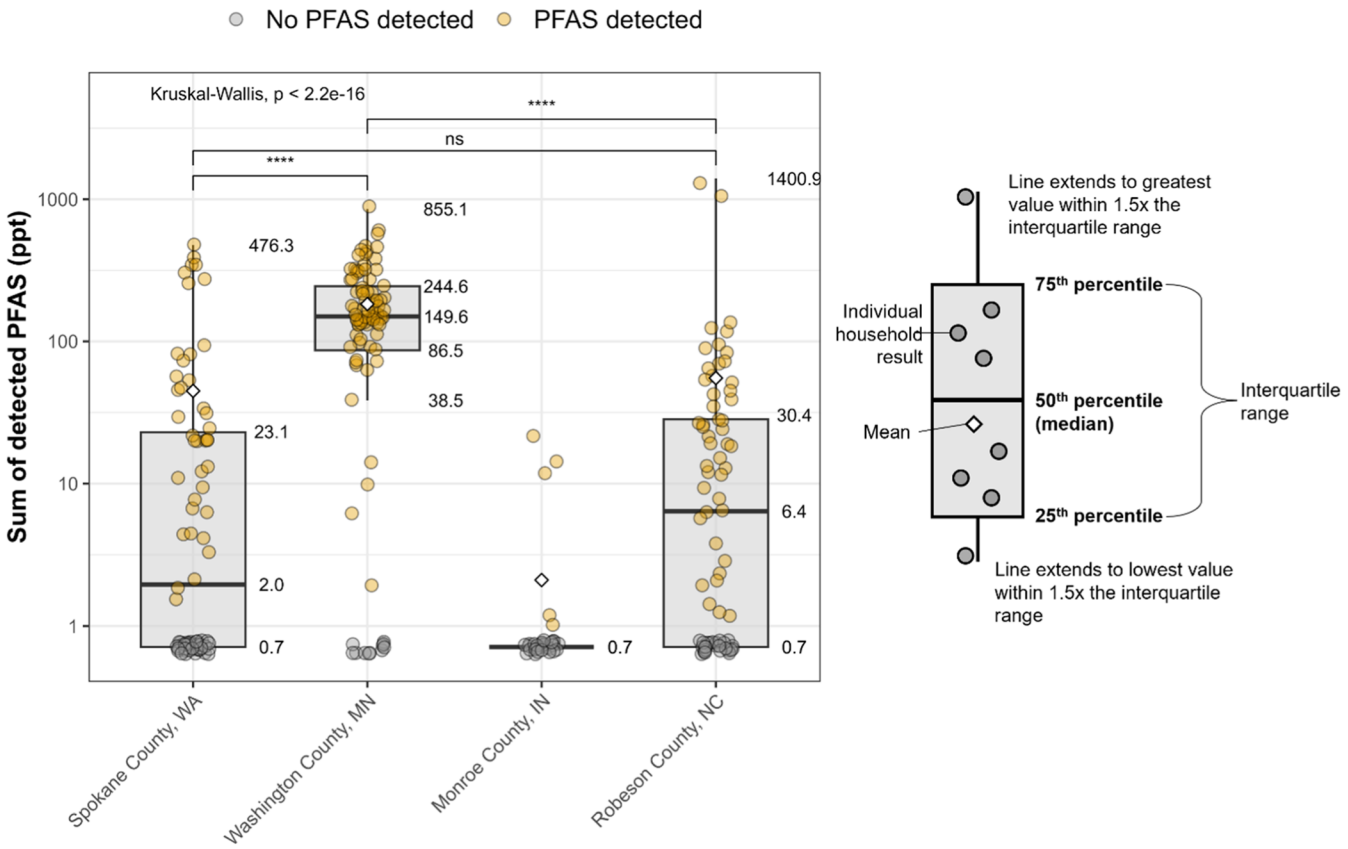


Figure 2. Distribution of ∑PFAS in private wells at each case study site. Each point in the figure represents an individual household. The points have been jittered slightly for visual clarity. Yellow points indicate that at least one individual analyte was detected above the MRL in the sample, while gray points indicate that no PFAS were detected above the MRL. Gray points are plotted at the average MRL of all of the individual analytes divided by the square root of 2. Pairwise comparisons are the result of Wilcoxon tests (ns = not significantly different at $p = 0.05$, **** = $p < 0.0001$). The Kruskal–Wallis test indicates the significance of the site on overall ∑PFAS concentrations. Numeric labels are shown for the 25th, 50th, and 75th percentile values and the maximum value within 1.5 times the interquartile range.

characteristics between all study participants and those completing the postparticipation survey.

To understand potential risk factors for PFAS in private well water, spatial regression models were run to analyze the relationships between the observed PFAS occurrence and distances to potential sources. Accounting for spatial effects was critical due to spatial autocorrelation in PFAS concentrations. Other recent studies using machine-learning approaches for predicting PFAS occurrence in drinking water sources (for example, recent work by Tokranov et al. using an extreme gradient-boosting model) have not accounted for spatial autocorrelation,[27] which could bias estimates of the effects of predictor variables on PFAS occurrence,[28] although for the Tokranov et al. study, the risk of bias was partially addressed by using a grid-based sampling approach. Bayesian spatial tobit models were fitted to analyze changes in the sum of all PFAS measured (natural log transformed) per unit distance (natural log of kilometers) from the nearest location of each source type. Bayesian spatial probit models were used to analyze relationships between the occurrence of total and individual PFAS above thresholds of interest (detection limits and relevant drinking water standards). The probit and tobit models were selected as an internally coherent approach to relate non-Gaussian (truncated and binary) data to the Gaussian scale. All models controlled for median household income and population density by census block group and for the study site due to prior research finding significant associations between these variables and the occurrence of PFAS in drinking water.[27,29] Main models included one source category (due to high correlations among locations of certain source types). In the sensitivity analyses, additional models representing multiple sources were tested.

RESULTS

In total, 271 households participated: 91 in MN, 74 in WA, 72 in NC, and 34 in IN. Most participants (81%) who answered demographic questions identified as White and lived in single-family homes (82%) (Table S4). In the racially diverse area of Robeson County, NC, however, all households that returned the survey identified as either American Indian or Black, and most (76%) resided in mobile or manufactured homes. Among all respondents, 18% had a high school degree or less, 50% had a Bachelor's or Associate's degree, and 31% held a graduate degree. Well characteristics also varied (Table S4), with depths ranging from less than 30 feet to over 500 feet. Well construction dates were relatively equally distributed by decade in the last 50 years. On average, participants obtained 86.0% of their drinking water and 96.1% of their cooking water from their wells (Table S5).

Prevalence of PFAS in Well Water. *Total PFAS.* Overall, 63% of the 271 samples had detectable PFAS. PFAS were found at each study site, with prevalences ranging from 15% (in Monroe County, IN) to 88% (Washington County, MN). Spokane County, WA, and Robeson County, NC, had 53 and

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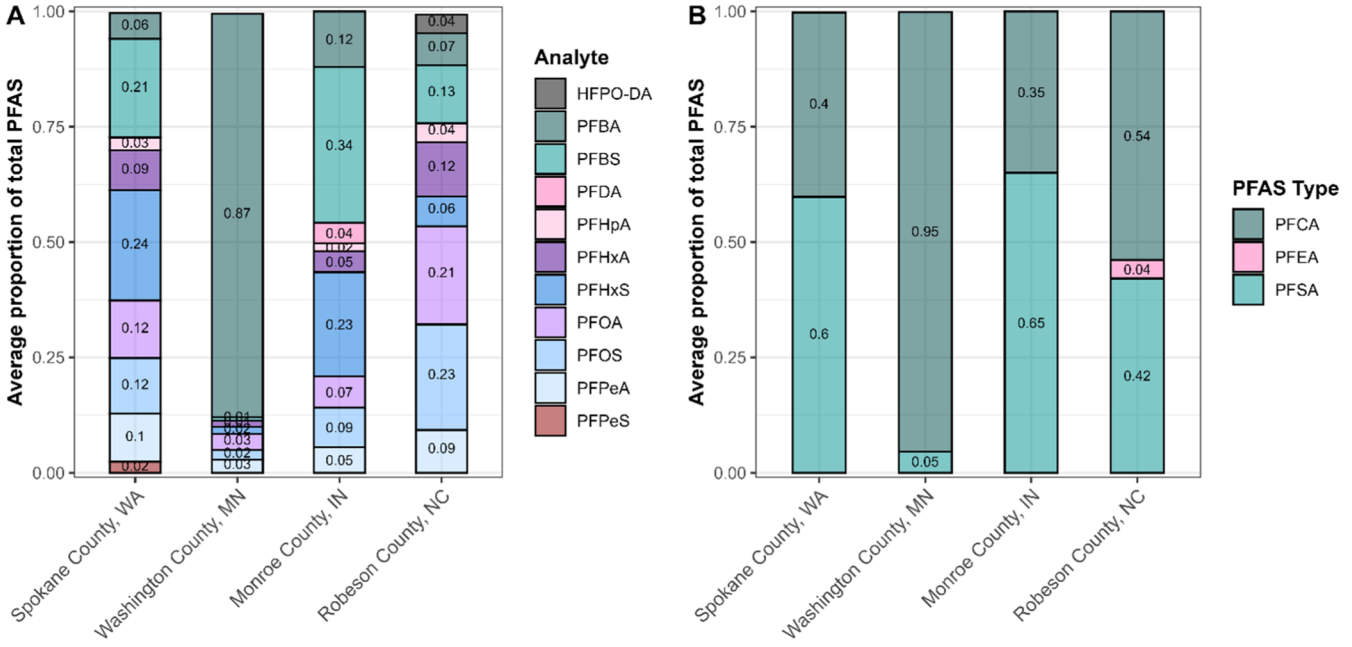


Figure 3. Average proportion of the total PFAS detected made up of individual PFAS analytes (panel A) and PFAS types (panel B) among wells where any PFAS were detected at each site. Wells where all analytes were < MRL are not included in the calculation of the average relative proportion. Individual analytes or types that made up less than 0.5% of the total PFAS on average for each site are not shown for visual clarity. Clear differences in the relative abundance of PFAS present at each site are evident due to different historical and ongoing PFAS releases in each area.

Table 1. Summary of PFAS Exceedances in Private Well Water Above Federal Health-Based Guidelines

	percent above an EPA maximum contaminant level goal (MCLG)[a]			percent above EPA maximum contaminant level (MCL)[b,c]				
case study site	at least one	PFOA	PFOS	PFOA	PFOS	HFPO−DA	PFHxS	Hazard Index
Spokane County, WA	42%	38%	38%	16%	19%	0%	19%	19%
Washington County, MN	60%	60%	43%	43%	19%	0%	8%	8%
Monroe County, IN	6%	6%	6%	0%	3%	0%	0%	0%
Robeson County, NC	61%	54%	57%	33%	31%	1%	0%	1%
All sites	49%	46%	41%	28%	20%	0.4%	8%	8%

[a]Any detection above the study MRL (1 ppt) was considered an exceedance above the MCLG of 0 for PFOA or PFOS. [b]Maximum contaminant levels (MCL) as of April 2024: 4 ppt for PFOA and PFOS, 10 ppt for HFPO−DA, PFHxS, and PFNA. The Hazard Index (HI) is the sum of the hazard quotients for HFPO−DA, PFBS, PFNA, and PFHxS based on their respective health-based limits, as follows: $\left(\frac{[\text{HFPO}-\text{DA}_{\text{ppt}}]}{[10\text{ ppt}]}\right)+\left(\frac{[\text{PFBS}_{\text{ppt}}]}{[2000\text{ ppt}]}\right)+\left(\frac{[\text{PFNA}_{\text{ppt}}]}{[10\text{ ppt}]}\right)+\left(\frac{[\text{PFHxS}_{\text{ppt}}]}{[10\text{ ppt}]}\right)=1$. [c]Maximum contaminant level goals (MCLG) = MCL for HFPO−DA, PFHxS, PFNA, and Hazard Index. No wells exceeded the EPA MCL of 10 ppt for PFNA.

64% detection rates, respectively. Adding together any results above the MRL for each of the 25 PFAS compounds ($\sum$PFAS$_{25}$), the two highest concentrations detected were from wells in Robeson County, NC, with 1401 and 1033 ppt. Median $\sum$PFAS$_{25}$ concentrations at the WA, MN, and NC sites (the three sites with known PFAS sources) were significantly higher than in Monroe County, IN, where there were no known point sources (Wilcoxon $p < 0.0001$; Figure 2). Additionally, among the three sites with known sources, Washington County, MN, showed significantly higher median $\sum$PFAS$_{25}$ concentrations than the WA and NC sites (Wilcoxon $p < 0.0001$; Figure 2). The median $\sum$PFAS$_{25}$ in Washington County, MN, was 183 ppt compared to 2.0 and 6.4 ppt in Spokane County, WA, and Robeson County, NC, respectively. There was no significant difference between the median $\sum$PFAS$_{25}$ values in Spokane County and Robeson County (Wilcoxon $p = 0.34$). In Monroe County, IN, the mean $\sum$PFAS$_{25}$ was 2.1 ppt, and the median was below the MRL. For the summary statistics (Table S6), results below the MRL were replaced with the MRL of the individual analyte divided by the square root of 2, with the MRL for $\sum$PFAS$_{25}$ considered to be the average of the MRLs of each of the analytes.

Individual PFAS. The most frequently detected PFAS and concentration distributions of individual PFAS varied by state (Figures S1 and S2). Wells in Spokane County, WA, showed greater detections of perfluoroalkyl sulfonic acids (PFSAs) than perfluoroalkyl carboxylic acids (PFCAs), with PFHxS and PFBS appearing in 43 and 47% of wells. In contrast, PFCAs were more frequently detected in Washington County, MN, where PFBA and PFPeA were detected in 88 and 81% of wells, respectively. In NC, a range of both PFSAs and PFCAs and long- and short-chain PFAS were detected. HFPO−DA was detected in 8% of NC samples but not in other locations. Lastly, Monroe County, IN, had very few households with any PFAS detections. The most frequently detected PFAS in Monroe County were PFBS, appearing in 9% of household wells, and PFHxS in 6%. PFOA and PFOS detection

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frequencies ranged from 6% in samples from IN for both to 60% in MN for PFOA. PFNA was less frequently detected, in 12% of samples from NC, 1% from WA, and none from MN or IN. Distributions of individual PFAS concentrations differed significantly by site (Figure S1), except for perfluorodecanoic acid (PFDA) and perfluoroheptanesulfonic acid (PFHpS); both long-chain PFAS that were infrequently detected had mean concentrations below the detection limit at all sites.

Relative Proportions of PFAS Types by Site. Wells in Washington County, MN, were dominated by a single analyte (PFBA), while the WA, IN, and NC sites showed a greater diversity of PFAS (Figure 3). PFBA made up 87% of the total PFAS detected on average in MN wells but only 7−12% at the other sites (Figure 3A). Overall, the broader class of PFCAs made up 95% of the detectable PFAS in Washington County, MN, compared to 35−54% at the other sites (Figure 3B).

PFAS Concentrations Compared to Health-Based Reference Values. Overall, 49% of wells exceeded the EPA MCLG for at least one PFAS (Table 1). This included results with any PFOA or PFOS detections as the MCLG is set at 0. EPA's new maximum contaminant level (MCL) of 4 ppt for PFOA and PFOS was also exceeded in 28 and 20% of wells tested, respectively. This proportion varied by state, with 43, 33, 16, and 0% above the PFOA MCL in MN, NC, WA, and IN, respectively. The MCL was set at the MCLG for HFPO−DA (10 ppt), PFHxS (10 ppt), PFNA (10 ppt), and the PFAS Mixtures Hazard Index (1).[6] HFPO−DA was above this level in only 1% of the samples. PFHxS was above in 8% of samples, only in wells in WA and MN. No wells exceeded 10 ppt for PFNA. Notably, while the overall levels of PFAS were highest in Washington County, MN, Spokane County, WA, showed the highest risk of exposure to PFAS mixtures according to the EPA's hazard index, with 19% of wells exceeding a hazard index of 1 (Table 1). The PFAS Mixture Hazard Index is calculated using the measured concentrations of HFPO−DA, PFBS, PFNA, and PFHxS, dividing each measured concentration by the corresponding health-based water concentration and then summing the resulting component hazard quotients.

Associations with Potential PFAS Source Types. Several potential PFAS source types included in the EPA PFAS Analytic Tools database were significantly associated with the total PFAS measured in private well samples based on the distance from each well to the nearest source of each type. Total PFAS concentrations were significantly higher in private wells closer to PFAS production facilities, Superfund sites, spill sites, and federal facilities (Figure 4). For example, the log of total measured PFAS increased by a factor of 3.15 per natural log unit decrease in the distance from a PFAS production facility. This translates to nearly a 9-fold increase in total PFAS concentration for a private well located 0.5 km from a production facility, compared to a well located 1 km from such a facility. Concentrations increased most as the distance to production facilities decreased, followed by Superfund sites, spill sites, and federal facilities. Estimation of the effect of distance from source on total PFAS concentrations remained robust in models accounting for multiple sources (Table S7). Selected sources were also significantly associated with the occurrence of PFOA, PFOS, and PFHxS at concentrations above the individual MCLs for those contaminants or with a PFAS hazard index above 1 (Figure 5). Sites with reported PFAS spills were consistently significantly associated with all of these indicators. Superfund sites and federal facilities were associated with PFOA or PFHxS occurrences above the MCL.

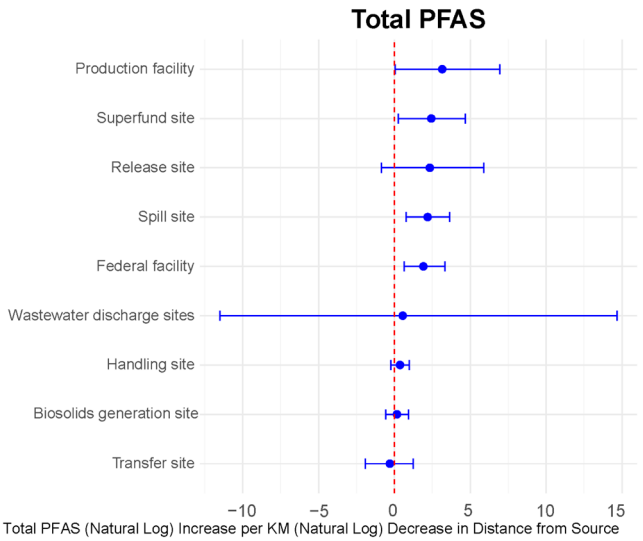


Figure 4. Change in total PFAS concentration (natural log) with inverse distance (natural log) from potential PFAS sources recorded in the EPA PFAS Analytical Tools Database and additionally compiled data sets. Potential source types are described in Table S2. Dots represent expected values of regression model coefficients, and whiskers are coefficients with 95% credible intervals. Values to the right of the dashed red line indicate that the total PFAS concentration increases as the distance to the indicated source decreases (equivalently, as the inverse distance increases).

Overall, the spatial regression coefficients shown in Figures 4 and 5 can be used to estimate the distance over which sources included in the EPA PFAS Analytical Tools database are associated with PFAS occurrence. Notably, distances to wastewater discharges, handling sites, locations where biosolids are produced, and transfer sites were not significant in any of the models. In models including just one source, toxic release inventory sites are significant predictors of whether the EPA hazard index exceeds 1.0 (β = 2.77, 95% CI 0.83−5.46) but not of any of the other outcomes. Release sites could not be included in models with multiple sources because their locations are highly correlated with those of production sites.

Effects of Study on Participant Actions. The post-participation survey (SI Appendix 4) was completed by 69 people (25% of all participants), well-distributed across WA, MN, and IN but including only three responses from NC. There were no significant differences between those participating in the study enrollment survey and those responding to the poststudy survey along key demographic indicators (race, income, education, and home ownership) (Tables S8 and S9).

Actions Recommended and Taken. Results reports (SI Appendix 3) included a recommendation to take action for 57% (155/271) of all participants, based on exceeding at least one health-based guideline or $\sum PFAS_{25}$ > 10 ppt. Among respondents, 51% (35/69) reported taking some action in response to their water test results (Table 2). Point-of-use filters were the primary recommended action. A recommendation to start using a filter significantly increased filter usage ($\chi 2$ (1, n = 63) = 15.1, p < 0.001). Among those receiving a recommendation to install a filter, 49% did so, compared to 0% of participants whose reports did not recommend a filter (because PFAS concentrations were below the guidelines).

Participants could select multiple actions in response to their test results (Table 2). The most common actions reported were "Started using water from the well for cooking" (32%),

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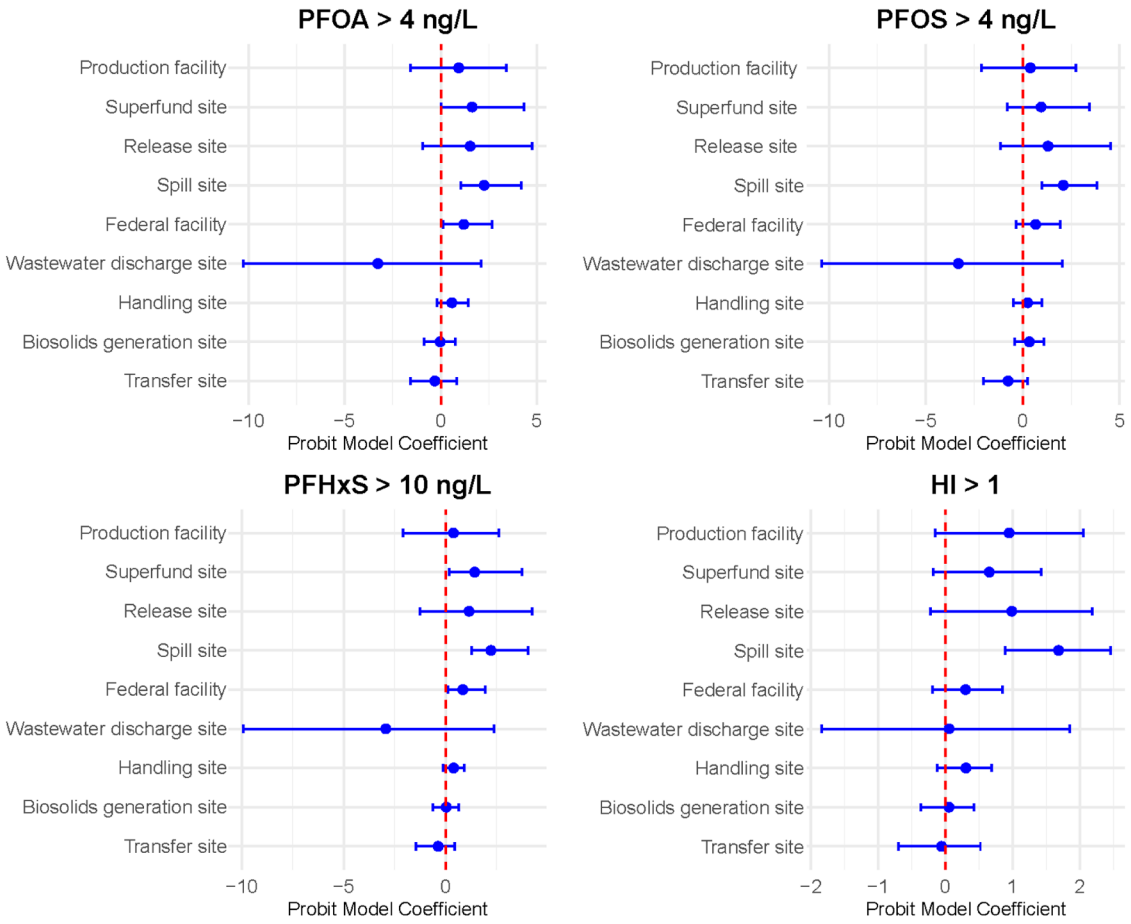


Figure 5. Relationship between the probability of detecting PFAS at concentrations above the EPA maximum contaminant level (MCL) and inverse distance (natural log) from potential PFAS sources recorded in the EPA PFAS Analytic Tools Database and additionally compiled data sets. Dots represent expected values of probit regression model coefficients, and whiskers are coefficients with 95% credible intervals. Values to the right of the dashed red line indicate that the probability of finding PFAS concentrations above the MCL increases as the distance to the indicated source decreases (equivalently, as the inverse distance increases).

Table 2. Postparticipation Survey Responses for Actions Taken in Response to Water Test Results, by whether any PFAS Were Detected and whether the Participant Report Included a Recommendation to Use a Filter

what actions did you take in response to the water test results?	with recommendation for action (n = 41)	without recommendation for action, with detections (n = 7)	without recommendation for action, without detections (n = 21)	all respondents (n = 69)
I started using a water filter (n = 64)	19/39 (49%)	0/6 (0%)	0/19 (0%)	19/64 (30%)
I stopped using a water filter (n = 61)	1/36 (2.8%)	0/6 (0%)	0/19 (0%)	1/61 (1.6%)
I chose a different type of water filter (n = 62)	11/37 (30%)	0/6 (0%)	0/19 (0%)	11/62 (18%)
I started drinking bottled water (n = 62)	4/37 (10.8%)	0/6 (0%)	1/19 (5.3%)	5/62 (8.1%)
I started drinking water from the well (n = 62)	13/37 (35.1%)	1/6 (16.7%)	5/19 (26.3%)	19/62 (30.6%)
I stopped drinking water from the well (n = 61)	2/36 (5.6%)	0/6 (0%)	1/19 (5.3%)	3/61 (4.9%)
I started using water from the well for cooking (n = 62)	14/37 (38%)	1/6 (16.7%)	5/19 (26%)	20/62 (32%)
I stopped using water from the well for cooking (n = 67)	2/36 (5.5%)	0/6 (0%)	0/19 (0%)	2/61 (3%)

"Started drinking water from the well" (31%), "Started using a water filter" (30%), and "Chose a different type of water filter" (18%). Of the 15 participants who began using well water after receiving results with an action recommendation, 12 added a filter or changed the type of filter. Among those who received a recommendation for action, those who adopted a filter were significantly more likely to start drinking their well water than those who did not (59 vs 15%, $X^2(1, n = 39) = 7.74$, $p = 0.005$).

Satisfaction with Study and Results' Communication. Participants demonstrated a high level of satisfaction with how the results were reported and with their overall engagement in the study (Figure 6). Of the 69 postparticipation survey responses, 93% were very satisfied with training, which was

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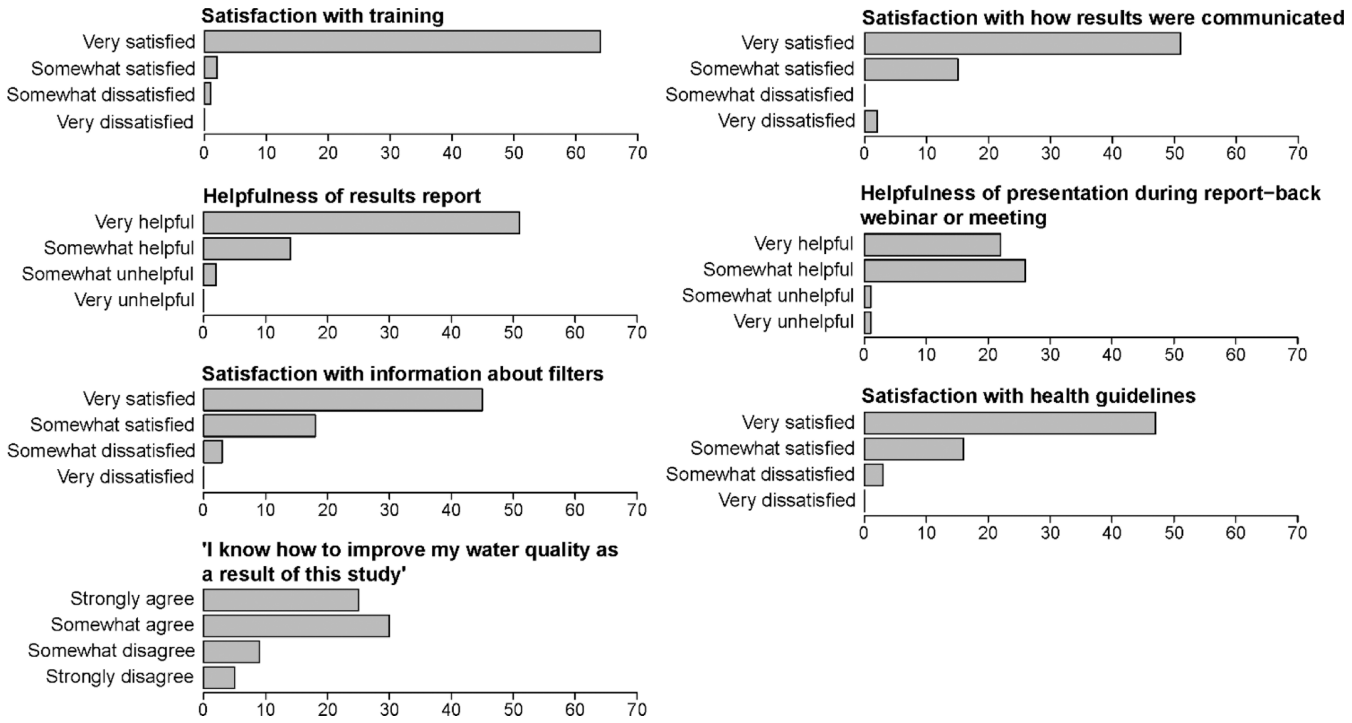


Figure 6. Postparticipation survey responses (n = 69) for satisfaction with how the results were reported and with their overall engagement in the study.

provided virtually, through written instructions, on-demand video demonstrations, and live webinars with Q&A. Ninety-five percent were somewhat or very satisfied with the information we shared about water filters and current guidelines. One commentor noted, "provided easy-to-follow guidelines and recommendations." Another said, "I appreciated the [state info] QR code and website info." Another mentioned the challenges of comparing this report to other information that used different reporting units (ppb vs ppt), and "I wish there had been more approved water filter recommendations." Some comments expressed frustration at the lack of established guidelines for all chemicals tested, but most were satisfied with the provided guideline information from federal and state sources. Most respondents (80% or 55/69) agreed that they know how to improve their water quality as a result of this study. Disagreement with this statement could have been from confusion about how to improve water quality or from a perception that their water quality did not need improving.

DISCUSSION

This study aimed to address gaps in knowledge of regional differences in PFAS prevalence in private wells, the influence of various sources on PFAS occurrence in private wells, and private well user decision-making in response to PFAS water testing results. Similar to our study, others have found PFAS in private well water in areas with no known releases.[11,24,30] Across the four dissimilar locations in our study, 63% of samples had detections of at least one PFAS above the study MRL. In contrast, other recent studies have found much lower overall PFAS detection rates in private wells of around 20%,[8−10] which may be a function of direct injection analysis, higher detection limits, as well as the targeted sampling in this study near known contaminated sites. A study of shallow wells throughout Wisconsin, however, detected PFAS in 71% of private wells tested,[31] and another study in Virginia detected PFAS in 95% of wells.[11]

Results above a health-based guideline in our study were uncommon in the IN county without a known source but still occurred. Our results are similar to those of another recent study investigating PFAS in 103 private well water samples in another rural IN county,[32] which found median total PFAS concentrations of 1 ppt, compared to 0.7 ppt in Monroe County, IN, in our study. In addition, the latter study found that none of the wells had PFAS levels above any MCLs, while in our study, 3% of IN wells exceeded PFOA MCLs.

Across all four sites, the most common results above an MCL were for PFOA. Rates of detection of legacy PFOA and PFOS above the MCL varied across our study areas, from a high of 43% for PFOA in MN to none in IN, and for PFOS, a high of 31% in MN and a low of 3% in IN. Other recent studies have found lower rates of an exceedance of this new standard, further highlighting the influence of local histories of PFAS use and disposal on PFAS risk in well water. In Florida, of 58 private well samples collected statewide, only 1 (2%) had PFOA and 3 (5%) had PFOS above the MCL.[33] In 450 samples from shallow private wells in Wisconsin, 17 (4%) had PFOS or PFOA above the MCL.[31] In nationwide tap water samples, including both private well and public-supply locations, the MCL for PFOA was exceeded in 6.7% and in 4.2% for PFOS, which may be a conservative estimate since the MCL was below the laboratory detection limit for some samples.[8] In studies in Santa Cruz County, Arizona, and Gustavus, Alaska, rates may have been higher compared with the new standard since 7.5% of wells tested in the former and 12% in the latter had results above the 70 ppt Public Health Advisory at the time for combined PFOA and PFOS concentration, with significant differences between wet and dry season measurements in Arizona.[30,34]

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Environ. Sci. Technol. 2025, 59, 16852−16863

Wells in Washington County, MN, were dominated by PFCAs, particularly PFBA. Overall, the broader class of PFCAs is generally associated with fluoropolymer manufacturing.[35] PFBA has been identified by the Minnesota Department of Health as one of the primary PFAS of concern in the area related to industrial fluorochemical manufacturing and waste disposal.[36]

The greater relative diversity of PFAS at the WA, IN, and NC sites may be a function of several interrelated factors, including different histories of PFAS releases, influences from multiple "overlapping" sources, and/or different geochemical sorption and transformation processes in each area.[37] As a control group without any known point sources in the area, the composition of PFAS in Monroe County, IN, may approximate the proportional background contributions from diffuse sources such as septic systems, atmospheric deposition from distant manufacturing sites, agricultural biosolids applications, and runoff from urban areas.[31,38−40] Notably, although the overall PFAS levels in Monroe County were much lower, the relative proportions of the detectable PFAS present were similar to those in Spokane County, WA, and Robeson County, NC. The wells sampled at these sites thus may be impacted by multiple complex source contributions compared to the single dominant source type in Washington County, MN (i.e., fluorochemical manufacturing waste). This may also be a function of the geographic distribution of wells around each known source (which was based on self-selection by study participants) and local geochemical differences affecting fate and transport in each area as well as differences in well construction and depth. Taking the NC site as an example, the two wells with the highest PFAS in NC (1401 and 1034 ppt) were in agricultural areas over 20 miles away from the manufacturing site and not near any known PFAS point sources, indicating the presence of additional unmapped PFAS risks in the area, potentially including land application of biosolids and septic system effluent.[31] HFPO−DA, also commonly referred to as "GenX," which was manufactured as a replacement compound by the nearby PFAS manufacturing facility after the phase-out of long-chain PFOA and PFOS,[19,41] was detected in only 8% of NC wells, compared with 85% of samples of private well water within 6 miles of the facility.[42] In contrast, PFOA, PFAS, and PFBS were detected in over 50% of NC wells in our study, compared with 37% with PFOA and 24% PFOS detections in the closer households.[42] Other replacement PFEAs traced to the NC manufacturing facility, such as perfluoro-2-methoxypropanoic acid (PMPA),[24,43] are not included in the standard EPA 533 analytical method, thus could not be assessed. The great variation in a relatively small geographic area highlights the challenges of designating areas at a greater or lesser risk for PFAS in wells.

Associations with Potential PFAS Source Types. Our study contributes additional insight into how distance from private wells to different potential PFAS source types is associated with total PFAS concentrations and the likelihood of exceeding guidance levels. The source types with the largest effects on total PFAS concentrations were PFAS production facilities, Superfund sites, locations where chemical spills containing PFAS were reported to the National Response Center (the federal emergency call center for reporting pollution releases), and federal facilities. A recent nationwide study of 269 private well and 447 public-supply tap water samples also found that total PFAS concentrations (sum of the 44 tested) decreased with distance from probable sources although they did not test the relationship with different source types.[8] Similarly, a recent nationwide study including water samples from a variety of well types (private, public, and monitoring wells) found that distance to potential sources was a significant predictor of the occurrence of at least one of 24 individual PFAS measured in the samples although these authors also did not distinguish source types (grouping all sources in PFAS Analytic Tools into one variable).[27] In analysis of samples from a network of monitoring wells in the eastern United States, most source categories were significantly associated with PFAS detections, and distance to the nearest fire-training areas was among the top predictors of detections.[9] Statewide random forest modeling for the sum PFOA and PFOS concentrations in Colorado did not find a clear pattern of importance for specific PFAS sources although the top 10 variables for discriminatory power included ski resorts, airports, and AFFF spills (similar to the data set used here for potential PFAS spills).[44] Potential sources by industry code and confirmed sources of plastics/rubber and textile manufacturers were the highest contributors to the accuracy of predicting PFAS detections in classification random forest models of PFAS data from private wells in New Hampshire.[10] Land application of biosolids from municipal and other wastewater streams has been identified as a potential source of PFAS contamination to groundwater[31,45,46] although the aforementioned recent study in rural IN found that land-applied biosolids were not associated with PFAS in private wells.[32] While our study cannot be compared directly to these other studies due to differences in study design (such as random or grid-based selection of sampling locations or selection based on proximity to sources), this work nonetheless adds to the body of evidence on types of sources that may have the biggest impacts on private wells.

Effects of Study on Participant Actions. The vast majority of participants (97%) perceived the format of results reports for their well water tests, which included health-based guidelines and information on filter effectiveness, as helpful for decision-making. Empowerment of study participants and communities for evidence-based decision-making and action is seen by many as a key benefit of returning research results.[47] Over half of the survey respondents reported taking some action in response to the results reports. Other recent evaluations of environmental health report-back efforts for other contaminants have also reported a third or more of participants taking action or planning to take action in response to results[21,22,48,49] although in some cases the rates were lower.[50] Providing specific recommendations on filter technologies and their effectiveness may have contributed to high satisfaction with the information, increased filter usage, and the start of well water use for drinking. Interviews in a prior study revealed that some trusted filters more after seeing testing results before and after installation. Some were unaware of low-cost point-of-use filter options.[25] Our reports included unfiltered water results and information on filter effectiveness.

Few other studies have reported on an evaluation of participant results reports for PFAS in water. Given the high satisfaction ratings, our example may benefit others reporting PFAS in water results for private well users. Hohweiler et al. included their report example in Supporting Information, and other researchers or public health laboratories may provide examples upon request.[11] Participants in a Florida study appreciated receiving results (unsolicited feedback), but some

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Environ. Sci. Technol. 2025, 59, 16852−16863

expressed concern about what the data meant and a desire for additional options for action.[33] In a 2009 biomonitoring study, interviewed participants reported frustration with the lack of information about health risk levels,[51] a continuing challenge. As evidence of PFAS health effects grows, benchmarks for several PFAS in water are now available. Some participant comments in this study expressed frustration at the lack of established guidelines for all chemicals tested, but most were satisfied with the provided information on federal and state guidelines. This highlights the importance of providing interpretive information alongside laboratory results.

Limitations. An important limitation in understanding the effectiveness of our reports is the low survey response rate from North Carolina, both during enrollment and after results were shared. Most (n = 216, 80%) returned the enrollment survey except for 50 participants in NC, 4 in IN, and 1 in WA. Surveys could not be collected from most participants in NC due to cultural and technological barriers during study recruitment, representing a limitation of our study. Tomsho et al. also noted uneven participation in their study and suggested identifying and addressing barriers to engagement and increasing outreach to those with higher contaminant levels.[48] For future research, we recommend earlier identification of potential barriers and codeveloping alternatives with community partners, such as SMS notifications for results and recruiting additional staff to verbally collect survey responses during or shortly after sample collection.

A limitation of our analysis of the influence of sources on PFAS in private wells is that the biosolids sites represent locations where these materials are generated, not where they are spread on land. The PFAS Analytical Tools databases currently only include biosolids generation sites; more information is needed on land application areas to better understand this potential pathway of PFAS in the environment. The data sets in the EPA Analytic Tools can continue to be improved by further verification of where PFAS releases to the environment have occurred, from among the potential sources. In addition, information about the PFAS mass released would add substantial benefit. Additionally, tools to identify sites that are listed under multiple source types could help further our understanding of the relative importance of different source types in terms of risks of PFAS occurrence in drinking water wells.

Our analysis did not consider transport pathway characteristics (such as hydraulic distances from sources to sampling sites) and transport times. However, we controlled for transport pathway effects by using a spatial regression approach. Spatial regression models account for transport pathway effects by capturing unmeasured or omitted spatial processes when detailed data on complex hydrologic pathways are unavailable.[52] Our analysis also did not control for all types of land use variables that have been associated with PFAS detections and concentrations in other recent studies.[8−10,19,27,31,44] Spatial regression also helps to control for such missing variables. In addition, recent modeling with a comprehensive data set of variables and long-term monitoring of wells in Minnesota found contamination sources playing a more significant role than land use or hydrogeologic characteristics in predicting PFAS exposure risks in groundwater.[53]

Recommendations. The drinking water regulations recently established for PFAS do not apply to unregulated private wells. Additional support is needed for private well users to understand and take action around the risks of PFAS in drinking water. Private well users can benefit from greatly expanded testing of water for PFAS, financial resources to prevent exposure, and additional guidance on household water filter selection and use. Advances in testing methods that can lower the cost will be important. Modeling and mapping approaches like those presented here, and those used in other recent studies, using nationwide data sets of potential sources, can help prioritize areas for testing, for example, in areas around potential PFAS spills and releases reported to the U.S. National Response Center. Hu et al. note that the correct identification of sources was important for model performance.[10] Detailed information on potential sources at a nationwide scale, like those in the EPA PFAS Analytic Tools database, can be used to estimate higher risk areas for private drinking water well contamination by PFAS above current guidelines and can help guide where to apply limited funds for testing and intervention.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c02521.

Additional detail on methods, table of complete survey results, regression tables for multisource models, examples of participant surveys and results reports, figure with distribution of individual PFAS analytes at each study side, and figure with proportion of private wells at each case study site with any detection of individual PFAS analytes (PDF)

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Notes

The authors declare no competing financial interest.

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